MIL-R-24169B(SH)
22 February 1982
SUPERSEDING
MIL-R-24169A(SHIPS)
22 March 1967
(See 6.5)

MILITARY SPECIFICATION

REGULATOR, AIR, DEMAND, SINGLE HOSE, DIVER'S

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a single hose, demand type regulator for use with self contained underwater breathing apparatus that will provide a diver with sufficient air on inhalation and will stop flow of air at the end of inhalation.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	-	Box, Folding, Paperboard.
PPP-B-636	-	Box, Shipping, Fiberboard.
PPP-B-640	-	Box, Fiberboard, Corrugated, Triple-wall.
PPP-B-676	-	Boxes Setup.
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-I-45208	-	Inspection System Requirements.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4220

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STANDARDS

FEDERAL

FED-STD-H28/2 - Screw-Thread Standards for Federal Services Section 2 Unified Thread Form and Thread Series for Bolts, Screws, Nuts, Tapped Holes and General Applications.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-889 - Dissimilar Metals.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Y14.36 - Surface Texture Symbols (DoD adopted)

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

COMPRESSED GAS ASSOCIATION, INC. (CGA)

Standard V-1; Compressed Gas Cylinder Valve Outlet and Inlet Connections

(Application for copies should be addressed to the Compressed Gas Association, Inc., 500 5th Avenue, New York, NY 10036.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.3).

3.2 Material. Unless otherwise specified herein, rigid pieces shall be made from metals suitable to the intended purpose (see 3.11).

3.2.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 Dissimilar metals. Dissimilar metals as defined by MIL-STD-889 shall not be used in the demand regulator assembly (see 3.6).

3.4 Screw threads. Screw threads shall be in accordance with FED-STD-H28/2.

3.5 Surface roughness. Surface finishes of working parts shall be in accordance with ANSI Y14.36. Surface finish of outside surfaces shall have a maximum roughness height rating of 125 prior to the application of any finish.

3.6 Demand regulator assembly. Demand regulator assembly shall be two stage design. Reference to the demand regulator assembly or regulator assembly shall be understood to include the first stage (high pressure) regulator, the connecting hose and the demand regulator, including all subassemblies and affixtures thereto. The first stage or high pressure regulator shall be back-mounted, that is, affixed to the high pressure cylinder manifold which is attached to the high pressure air cylinder(s) carried by the diver on his back. The first stage regulator shall be mounted to extend from the manifold not more than 3 inches toward the diver (see figure 1). The second or low pressure stage, hereinafter referred to as the demand regulator, shall be connected to the high pressure stage by the hose of 3.13. The mouthpiece, exhaust valve(s), exhaust manifold, and demand regulator shall be a single integral unit.

3.6.1 Purge feature. Demand regulator shall include a feature that will allow a diver to depress the diaphragm of the regulator in order to cause a free flow, purging the mouthpiece of water.

3.7 Supply pressures. Demand regulator assembly shall function in accordance with this specification with supply pressures from 3,000 to 200 pounds per square inch (lb/in^2). The high pressure regulator shall withstand application of 5,000 lb/in^2 hydrostatically without sustaining damage (see 4.7.2).

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3.8 Continuous flow. Demand regulator assembly shall be designed so that regardless of the diver's position, continuous flow will not occur.

3.9 Leakage. Demand regulator assembly shall not leak (see 4.7.3).

3.10 Manifold connection. Demand regulator assembly shall attach to a high pressure (3000 lb/in^2) breathing air supply by means of a nipple and yoke arrangement in accordance with connection No. 1310 of CGA Standard V-1. The yoke shall accommodate both maximum and minimum valve outlet contours.

3.11 Parts exposed to pressure. Metal parts subjected to pressure shall be made of brass.

3.12 Exterior finish. Externally exposed parts shall have a dull black non-reflecting finish which shall neither crumble or flake. The finish of exposed metal parts shall be non-corroding. The finish shall be capable of being easily repaired by operators. Materials required to maintain and repair the finish shall be furnished, in small amounts, with the special tools of 3.21.

3.13 Hose. The hose connecting the high pressure regulator and the demand regulator shall be exceedingly flexible and shall be resistant to deterioration due to exposure to sunlight, salt water, oil or hydrocarbons. The hose shall have a safe working pressure of 250 lb/in^2.

3.13.1 Hose couplings. One of the hose couplings shall be a "swivel" type connection that will allow the alinement of the hose to be changed to eliminate torques that would otherwise be transferred to the mouth bit. The hose with couplings shall be between 24 and 26 inches long. The hose with couplings attached shall be capable of withstanding 250 lb/in^2 with a static tension of 50 pounds applied at the couplings.

3.14 High pressure filter. The air inlet of the first stage regulator shall have a sintered bronze filter. The filter porosity shall be determined on the basis of the valve opening of the demand regulator.

3.15 Protective cap. There shall be a protective cap of plastic or other material for the air inlet. The protective cap shall be secured to the yoke fitting by a flexible means to preclude loss of the cap when not in use.

3.16 Exhaust gas. Exhaust gases shall be vented in a manner such as to least inhibit the diver's vision.

3.17 Mechanical peak inhalation and exhalation pressures. Demand regulator assembly shall exhibit, at a maximum, the values of peak inhalation and exhalation pressures shown on figure 2 when tested on a mechanical breather in accordance with 4.7.1.1 in ambient pressures up to the equivalent of 200 feet of sea water.

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3.18 Subjective peak inhalation and exhalation pressures. Demand regulator assembly shall demonstrate, at a maximum, the values of peak inhalation and exhalation pressure shown on figure 2 when used by a typical U.S. Navy diver having a maximum tidal volume of 5 liters at simulated depths up to 200 feet of sea water swimming against a 12 pound force (see 4.7.1.2).

3.19 Steady flow inhalation and exhaust pressures. When tested in accordance with 4.7.4, the regulator assembly shall exhibit a steady flow inhalation pressure not to exceed 5 inches of water. When tested in accordance with 4.7.6, the steady flow exhaust pressure shall not exceed 4 inches of water.

3.19.1 Steady flow inhalation pressure variation with flow rate. The steady flow inhalation pressures required by the demand regulator assembly to attain various flow rates shall be not greater than the values shown on figure 3.

3.20 Flow rates. When tested in accordance with 4.7.5, the air flow rate at a reduced pressure of 200 $\pm$ 10 lb/in^2 shall be not less than 60 percent of the air flow rate at the initial air supply pressure.

3.21 Special tools. The design of the regulator shall be such that the need of special tools for adjustment and maintenance shall be kept at a minimum (see 3.12). Special tools for adjusting and maintaining the regulator assembly shall be provided as part of regulator assembly.

3.22 Technical data. The contractor shall prepare engineering drawings and technical manuals in accordance with the data ordering documents specified (see 6.2.2).

3.22.1 Technical manuals. In addition to the general requirements specified (see 6.2.2), the technical manuals shall include instructions for maintaining and repairing the black, non-reflecting exterior finish of the regulator.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208.

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4.1.2 Test conditions. All tests shall be conducted at a temperature of 70 ± 5°F.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Sampling.

4.3.1 Lot. All regulator assemblies offered for delivery at one time shall be considered a lot.

4.3.2 Sampling for visual and dimensional examination. From each lot of regulator assemblies, samples shall be selected in accordance with MIL-STD-105 at General Inspection Level III for the examination specified in 4.6. The AQL shall be 1.5 percent defective.

4.3.3 Sampling for tests other than visual and dimensional examination. From each lot of regulator assemblies, samples shall be selected in accordance with special inspection level S-4 of MIL-STD-105 for the tests specified in 4.5, other than visual and dimensional examination of 4.6.

4.4 First article inspection. Two regulator assemblies shall be subjected to the first article inspection shown in table I, except for the inhalation and exhalation pressure tests of 4.7.1.1 and 4.7.1.2 (see 4.4.1). The tests shall be conducted in the order shown in table I. The contractor shall prepare a test report in accordance with the data ordering document included in the contract (see 6.2.2).

TABLE I. Inspection.

Examination and test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection
Visual and dimensional examination	3.2, 3.3, 3.4, 3.5, 3.12	4.6	X	X
Inhalation and exhalation test pressure:				
mechanical peak	3.17	4.7.1.1	X	
subjective peak	3.18	4.7.1.2	X	
Hydrostatic	3.7	4.7.2	X	X

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TABLE I. Inspection. - Continued

Examination and test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection
Leakage	3.9	4.7.3	X	X
Steady flow inhalation pressure	3.19	4.7.4	X	X
Flow rate	3.20	4.7.5	X	X
Steady flow exhalation pressure	3.19	4.7.6	X	X

4.4.1 First article inspection at a Government facility. The first article sample of 4.4 shall be forwarded to a Government facility as specified (see 6.2.1 and 6.3.1) for the mechanical peak inhalation and exhalation pressure test of 4.7.1.1, and the subjective peak inhalation and exhalation pressure test of 4.7.1.2.

4.5 Quality conformance inspection. The sample units selected in accordance with 4.3.2 or 4.3.3 as appropriate shall be subjected to the quality conformance inspection tests shown in table I. If any regulator fails in any test (see 4.7), the entire lot represented by the sample shall be rejected. The contractor shall prepare a test report in accordance with the data ordering document included in the contract (see 6.2.2).

4.5.1 Quality conformance inspection reports. Quality conformance inspection reports shall be prepared in accordance with the data ordering document specified (see 6.2.2).

4.6 Visual and dimensional examination. Each of the sample regulator assemblies selected in accordance with 4.3.2 shall be examined to verify conformance with all of the requirements of this specification which do not involve tests. Any sample regulator having one or more defects shall be rejected. If the rejection number exceeds the acceptance number defectives, the lot represented by the sample shall be rejected.

4.7 Tests.

4.7.1 Inhalation and exhalation pressure.

4.7.1.1 Mechanical peak. The sample units shall be tested by a simulated breathing test using a breathing machine having tidal volume of 2 liters set at a rate of 20 breaths per minute. For a given ambient pressure, the maximum

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pressure differential observed between ambient pressure and pressure inside the mouthpiece during any exhalation cycle shall be the mechanical peak exhalation pressure (see 3.17).

4.7.1.2 Subjective peak. Sample units which have passed the testing of 4.7.1.1 shall be tested using the arrangement shown on figure 4. For a given depth, the maximum pressure differential observed between ambient pressure at mouthpiece level and pressure inside the mouthpiece during any inhalation cycle shall be the subjective peak inhalation pressure. For a given depth, the maximum pressure differential observed between ambient pressure at mouthpiece level and pressure inside the mouthpiece during any exhalation cycle shall be the subjective peak exhalation pressure (see 3.18).

4.7.2 Hydrostatic. The high pressure regulator of each sample shall be hydrostatically tested to a pressure of 5,000 lb/in^2 applied internally. The high pressure regulator shall not leak during this test (see 3.7).

4.7.3 Leakage. The sample regulator assembly shall be mounted in the test tank and shall be connected to the appropriate inlet and outlet air lines as shown on figure 5. The main throttling valve of figure 5 shall be closed and high pressure air of 3,000 $\pm$ 25 lb/in^2, shall then be supplied to the regulator assembly. The tank shall then be partially filled with water up to a level such that the assembly under test is submerged to a depth of 2 inches at the highest point. If there are signs of air leakage from the regulator assembly, the cause shall be determined and if found to be due to mismating with the test apparatus, it shall be eliminated. Low pressure air shall then be admitted into the top of the tank and the pressure raised to and maintained at 5 $\pm$ 0.2 lb/in^2 for all tests. Leakage at any time, other than that due to mismating with the test apparatus or ambient pressure decrease, shall be cause for failure of the leakage test (see 3.9).

4.7.4 Steady flow inhalation pressure. The main throttling valve shown on figure 5 shall be adjusted to a setting such that an air flow rate through the air outlet line, as indicated on the direct reading air flowmeter, of 6 $\pm$ 0.25 cubic feet (ft^3) of free air per minute is attained and sustained. Determinations shall be made under steady flow conditions of the pressure differential as indicated by the water manometer and the head of water above the center of the demand valve diaphragm as indicated by the tape rule mounted inside the tank, the sum of these pressures, in inches of water, being the Steady Flow Inhalation Pressure (see 3.19).

4.7.5 Flow rate. The throttling valve in the air outlet line as shown on figure 5 shall be opened slowly and shall be adjusted to a setting such that the air flow rate through the air outlet line, as indicated on the direct reading air flowmeter, is 8 $\pm$ 0.5 ft^3 of free air per minute. When steady flow conditions have been obtained, the total pressure differential on the sample shall be noted. With this valve held constant by suitable adjustment of the throttling valve in the air outlet line, the air supply pressure to the sample assembly shall be reduced from 3,000 to 200 $\pm$ 10 lb/in^2. When steady flow conditions have again been obtained, the flow rate will again be noted. This air flow rate shall be the air flow rate at reduced pressure (see 3.20).

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4.7.6 Steady flow exhalation pressure. The pressure differential necessary across the installed exhaust valve to sustain a flow through the exhaust valve of 6.00 ± 0.25 ft^3 per minute while exhausting to a minimum pressure of 0.083 lb/in^2 (or 2.25 inches of water) shall be determined. The demand (second stage) regulator shall be immersed to a minimum depth of 2 inches in water, secured in the attitude in which it will normally be held in the mouth of a diver while swimming. Low pressure air shall be supplied to the mouthpiece by a system which shall contain devices for measuring flow through the mouthpiece and pressure inside the mouthpiece. The pressure at the highest (shallowest) point of the highest port through which air passes to the surrounding water is the ambient exhaust pressure. By means of the low pressure air supply, a steady flow of 6.00 ± 0.25 ft^3/minute and air shall be sustained from the mouthpiece through the exhaust valve and the exhaust manifold to the surrounding water. The differential between the pressure shall be the Steady Flow Exhaust Pressure. To start the test, the exhaust manifold shall be filled with water (see 3.19).

4.8 Inspection of packaging. Preservation, packaging, packing, and marking shall be inspected to assure compliance with section 5 of this specification.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation and packaging.

5.1.1 Level A. Preservation and packaging shall be in accordance with MIL-P-116 as follows:

(a) Tools. Meeting the requirements of Method III, special tools (see 3.21) shall be wrapped and contained in a snug-fitting folding or set-up paperboard box conforming to the requirements of PPP-B-566 or PPP-B-676 at the option of the manufacturer. Box closure shall be in accordance with the appendix to the applicable box specification.

(b) Demand regulator assembly. Each demand regulator assembly (see 3.6) shall be packaged in a heat sealed plastic bag.

(c) One set of special tools, one demand regulator assembly and a manual (see 3.22.1) shall be cushioned and packaged in a snugfitting fiberboard box conforming to PPP-B-636. Box closure and sealing shall be in accordance with the box specification.

5.1.2 Level C. Special tools (see 3.21) including maintenance and repair items (see 3.12) shall be wrapped and enclosed in a snug-fitting box. This box, a demand regulator assembly (see 3.6) and a manual (see 3.22.1) shall be suitably cushioned and boxed in a single package in accordance with the suppliers best commercial practice sufficient to afford adequate protection against physical damage during shipment from the supply source to the using activity for immediate use.

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5.2 Packing.

5.2.1 Level A. Unless otherwise specified (see 6.2.1), equipment packaged shall be packed in fiberboard boxes conforming to PPP-B-636 (class weather resistant) or PPP-B-640 (class 2) at the option of the contractor. All center and edge seams and the manufacturers joints shall be sealed and waterproofed with pressure sensitive tape in accordance with the applicable fiberboard box specification. Box closure and banding shall be in accordance with the applicable box specification when specified. Unit fiberboard containers conforming to class weather resistant, closed, sealed and banded as specified herein, may be used as the shipping container and need not be overpacked.

5.2.2 Level B. Units of basic equipment, packaged as specified (see 6.2.1), shall be packed in fiberboard boxes conforming to PPP-B-636 (class domestic) or PPP-B-640 (class 1) at the option of contractor. Shipping containers shall be closed and banded in accordance with the applicable container specification, except that fiberboard containers may be sealed with tape conforming to PPP-T-45. When specified (see 6.2.1), unit containers closed, sealed and banded as specified herein may be used as the shipping container and need not be overpacked.

5.3 Marking. In addition to any special marking requirements, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The regulator covered by this specification is intended to supply a flow of life support air to the diver upon his inhalation and to stop the flow of air upon the end of inhalation.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a first article is required for inspection and approval (see 3.1, 4.4, and 6.3).
(c) If tests are required at a Government facility, indicate the name and address of facility (see 4.4.1 and 6.3.1).
(d) Preservation, packaging, packing, or marking requirements other than those required by 5.1 (see 5.2.1 and 5.2.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When

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the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirements title	Applicable DID no.	Option
3.22	Drawing, engineering and associated lists	DI-E-7031	Level 3
3.22.1	Manual, technical, standard basic issue	DI-M-2044	Type I of MIL-M-15071
4.4	Test report (first article inspection)	DI-T-2072	10.1b
4.5	Test report (quality conformance inspection)	DI-T-2072	10.1b

(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3.1 The acquisition originator should arrange with the Naval Sea Systems Command (Code 00C) for the specific facility to be used for the first article testing for the mechanical peak inhalation and exhalation pressure test of 4.7.1.1 and the subjective peak inhalation and exhalation pressure test of 4.7.1.2.

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6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy-SH
(Project 4220-N028)

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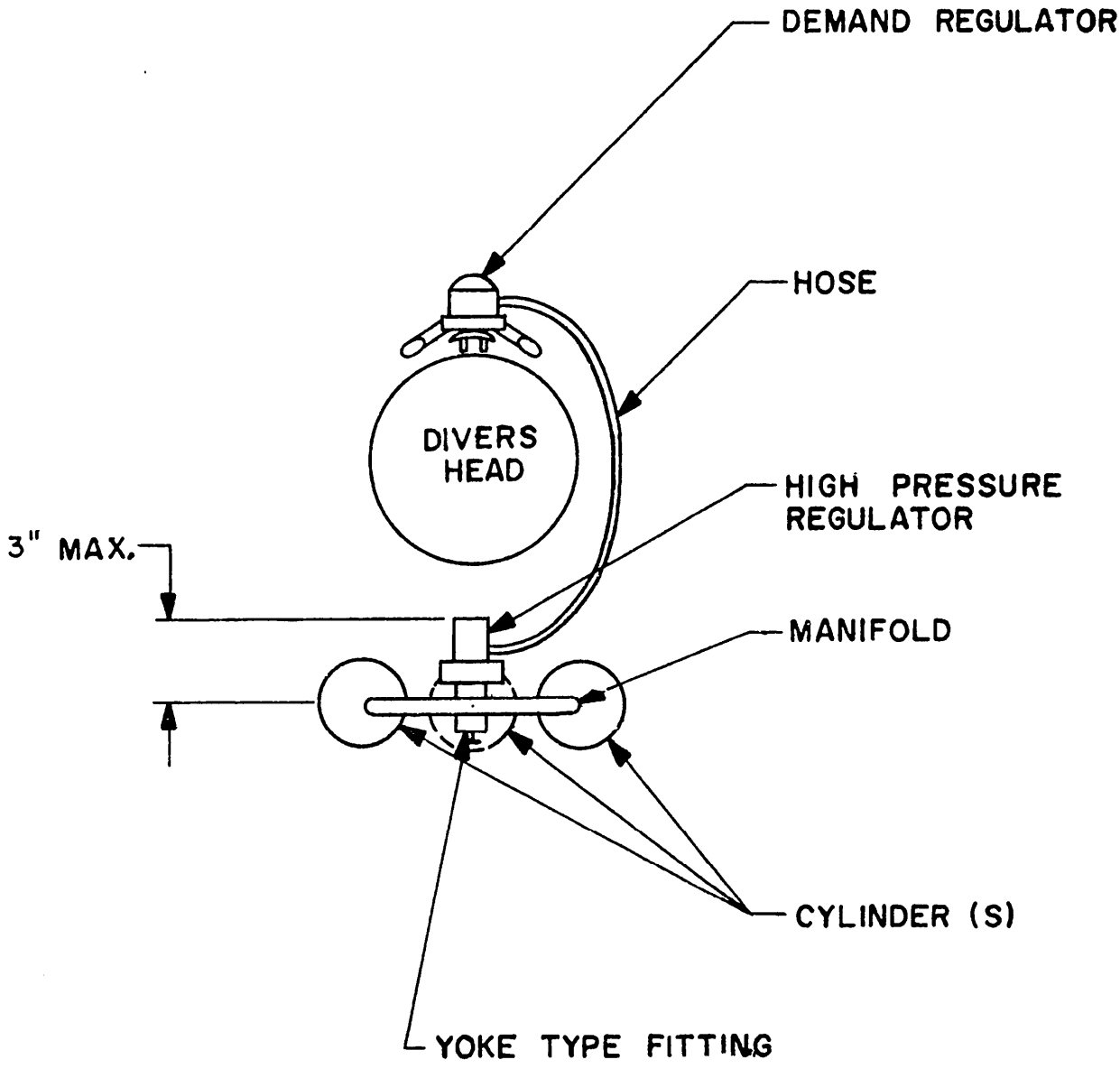
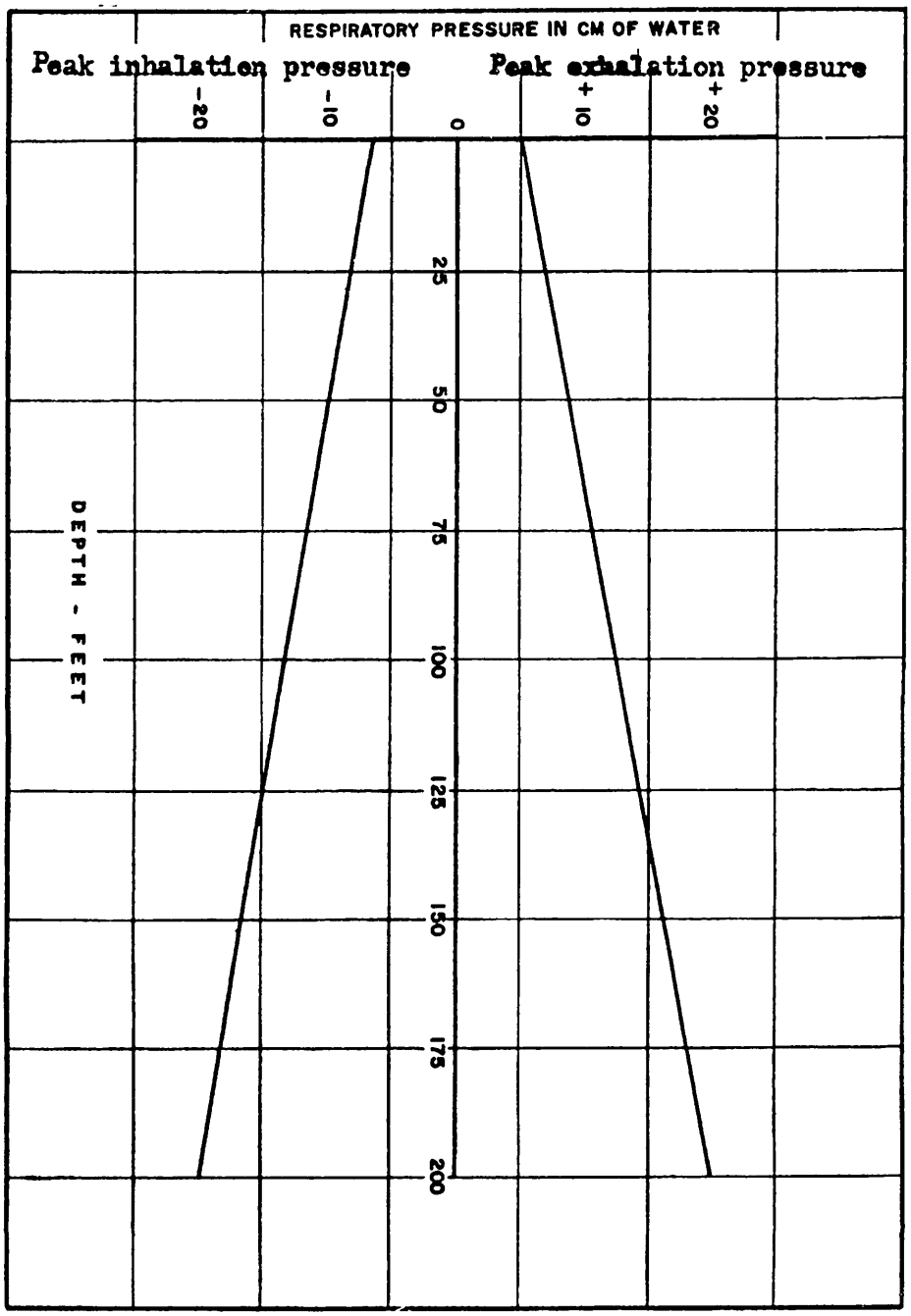


FIGURE 1. First stage regulator extension limitation.

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FIGURE 2. Respiratory pressures.

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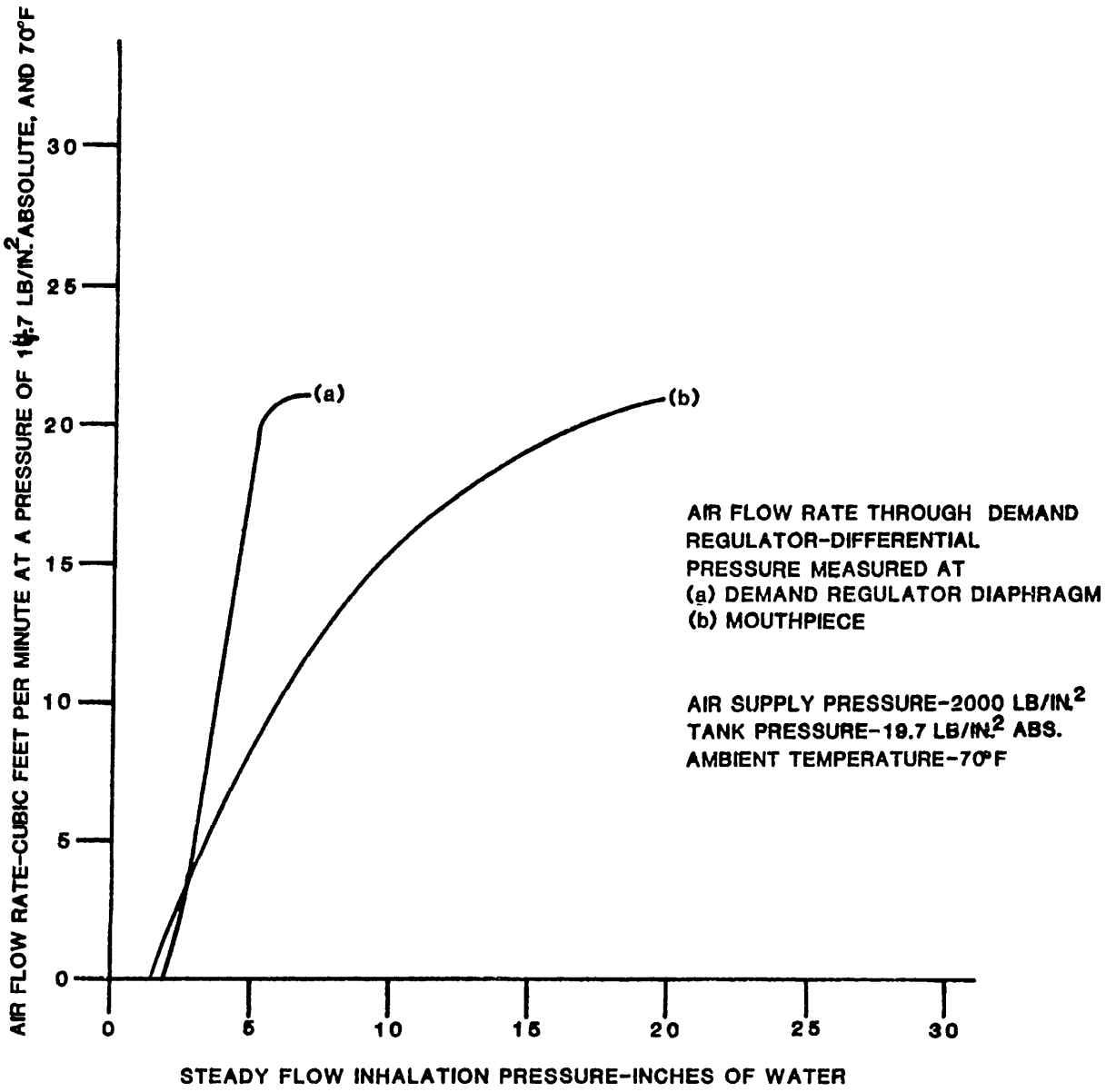


FIGURE 3. Flow rates.

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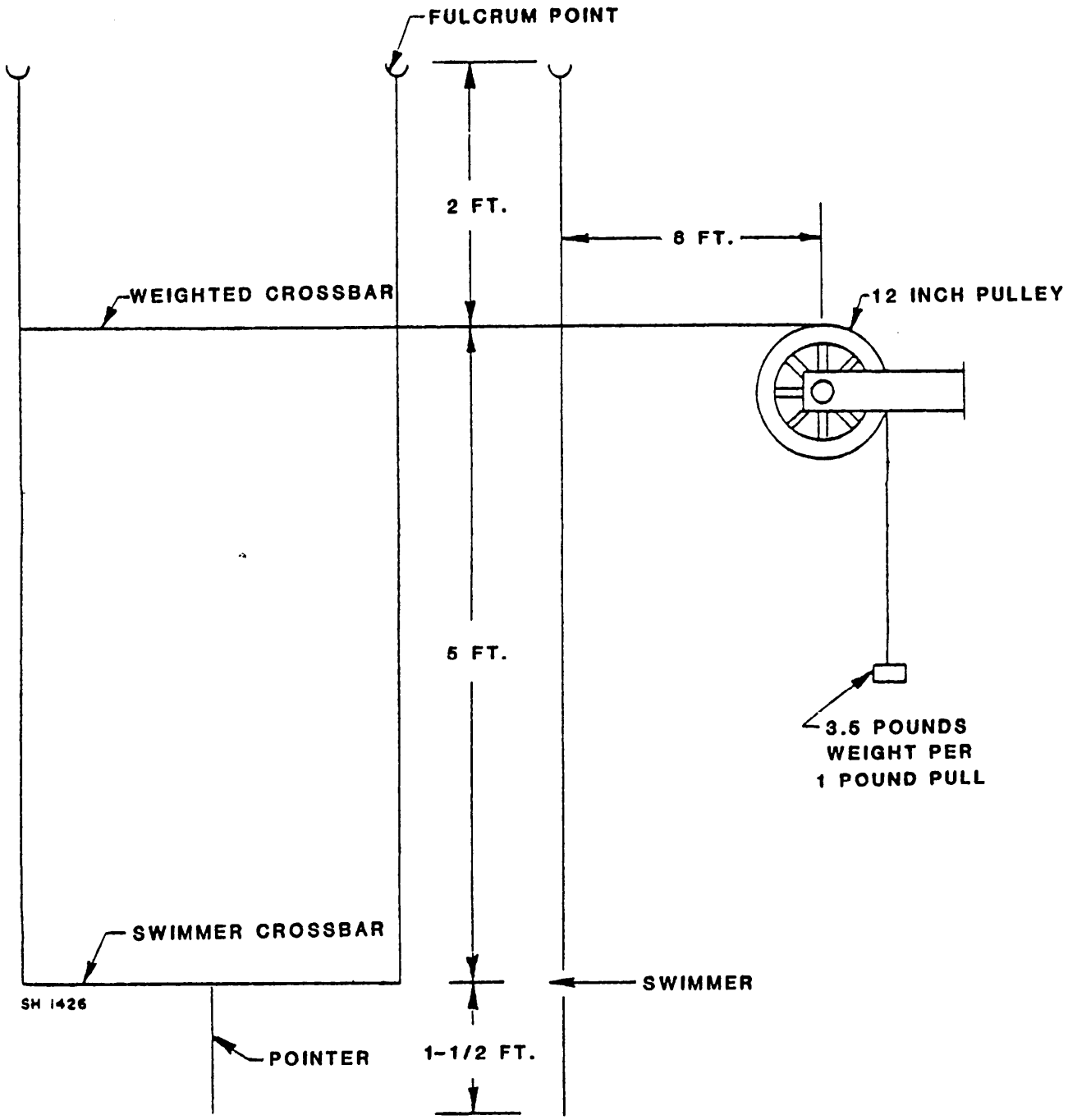
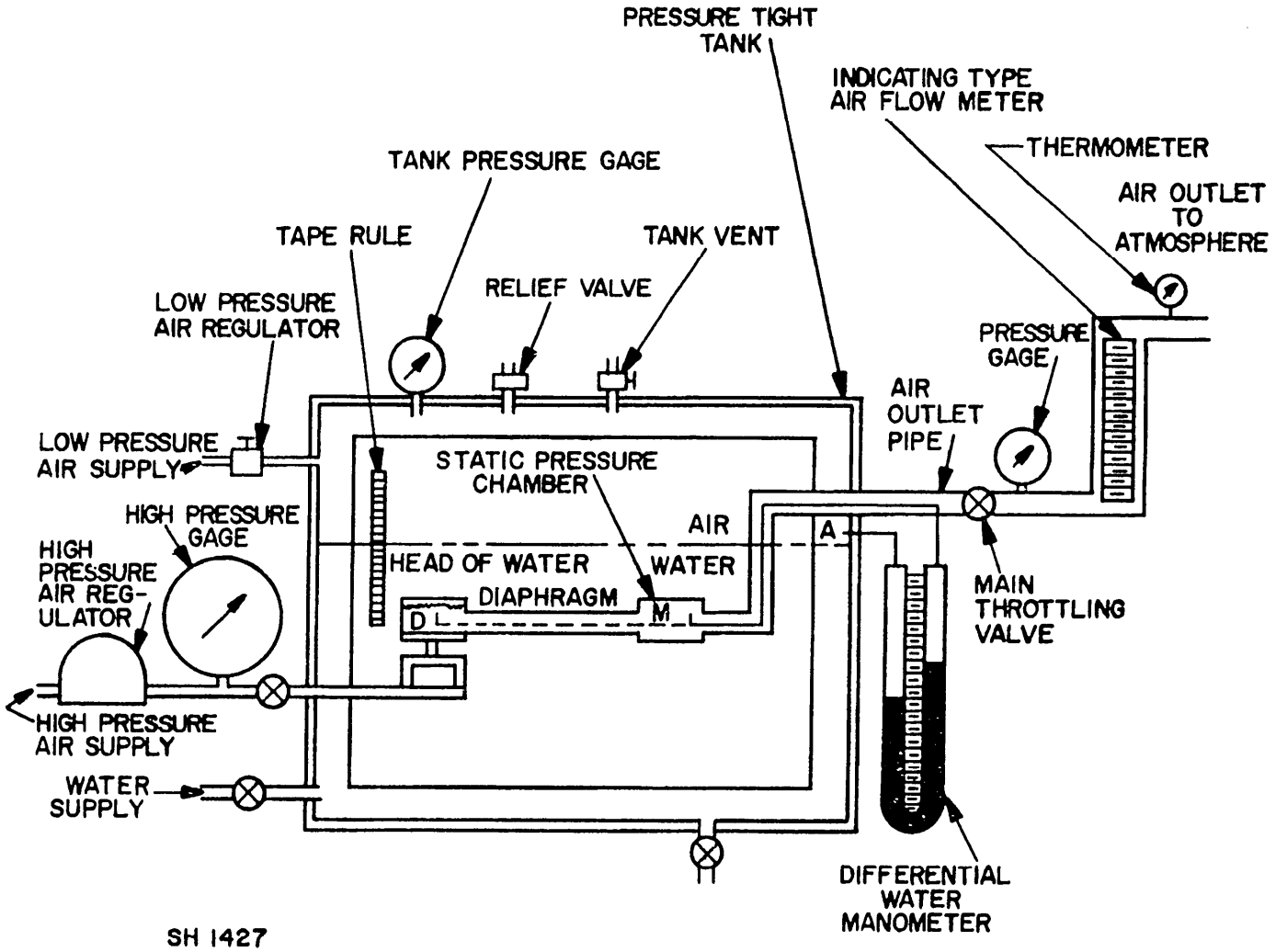


FIGURE 4. Simulated swimming test arrangement.

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A - PRESSURE TAP AT TANK TOP AIR SPACE.
D - PRESSURE TAP AT DEMAND VALVE DIAPHRAGM.
M - PRESSURE TAP AT DEMAND ASSEMBLY MOUTH-PIECE OR FACE MASK.

FIGURE 5. Test arrangement.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-R-24169B(SH)

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